CROSS PRODUCT & CURL (determinant)

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$(A_yB_z - A_zB_y)\hat{x} - (A_xB_z - A_zB_x)\hat{y} + (A_xB_y - A_yB_x)\hat{z}$$

$$\nabla\times\mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z row survives.

DOT PRODUCTS $\hat{a}\cdot\hat{b}$

Same dir = 1; perpendicular = 0.

	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$	$\hat{\theta}$
$\hat{x}$	1	0	0	0	0	0
$\hat{y}$	0	1	0	0	0	0
$\hat{z}$	0	0	1	0	0	0
$\hat{\phi}$	0	0	0	0	1	0

SURFACE ELEMENT $d\mathbf{s}$

$d\mathbf{s} = \hat{n}\,da$ *RHR: curl fingers in loop dir. $\Rightarrow$ thumb $= \hat{n}$*

Loop plane	$d\mathbf{s}$
x-y plane	$\hat{z}\,dx\,dy$ or $\hat{z}\,r\,dr\,d\phi$
y-z plane	$\hat{x}\,dy\,dz$
x-z plane	$\hat{y}\,dx\,dz$ (note $\hat{y}$, not $-\hat{y}$!)
toroid/coplanar	$\hat{\phi}\,dr\,dz$

Uniform $\mathbf{B}$*: $\iint da = A$ (just loop area). RHR picks sign*

B FIELDS

Infinite wire *single wire carrying I*

$$\mathbf{B} = \hat{\phi}\,\frac{\mu_0 I}{2\pi r}$$

Inside toroid *N-turn coil carrying I*

$$\mathbf{B} = \hat{\phi}\,\frac{\mu N I}{2\pi r}, \qquad \mu = \mu_r\mu_0$$

N in $\mathbf{B}$ only if source is multi-turn coil. Single wire: $N = 1$ (drop it). N in $V_{emf} = -N\,d\Phi/dt$ is the receiver *turns.*

$\partial B/\partial t$ TABLE

$B(t)$	$\partial B/\partial t$
$B_0\sin(\omega t)$	$\omega B_0\cos(\omega t)$
$B_0\cos(\omega t)$	$-\omega B_0\sin(\omega t)$
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
B_0 (const)	0

SIGN $\to$ CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	$- \to +$ terminal	$-\hat{\phi}$ (CW)
> 0	$+ \to -$ terminal	$+\hat{\phi}$ (CCW)
$= 0$	—	none

Lenz: induced I opposes the change in Φ.

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a \to b$

$$\int_a^b \frac{dr}{r} = \ln\frac{b}{a} = \ln b - \ln a$$

$\ln 2 = 0.693$; $\ln 3 = 1.099$; $\ln 1.2 = 0.182$; $\ln 1.5 = 0.405$

j IDENTITIES

Expr	Val	Expr	Val	Expr	Va
j^2	-1	$1/j$	$-j$	j^3	$-$
$e^{j\pi/2}$	j	$e^{-j\pi/2}$	$-j$	$e^{j\pi}$	$-$
$\lvert a+jb\rvert$	$\sqrt{a^2+b^2}$	$\angle(a+jb)$	$\arctan(b/a)$	$j\cdot(-j)$	1

$\mathrm{Re}\{Ae^{j\theta}e^{j\omega t}\} = A\cos(\omega t + \theta)$.

UNIT CIRCLE (KEY ANGLES)

θ	$\cos\theta$	$\sin\theta$	θ	$\cos\theta$	$\sin\theta$
0	1	0	$2\pi/3$	$-1/2$	$\sqrt{3}/2$
$\pi/6$	$\sqrt{3}/2$	$1/2$	$3\pi/4$	$-\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	π	-1	0
$\pi/3$	$1/2$	$\sqrt{3}/2$	$3\pi/2$	0	-1
$\pi/2$	0	1	2π	1	0

SI PREFIXES

Prefix	Sym	Val	Prefix	Sym	Val
pico	p	10^{-12}	kilo	k	10^{3}
nano	n	10^{-9}	mega	M	10^{6}
micro	μ	10^{-6}	giga	G	10^{9}
milli	m	10^{-3}	tera	T	10^{12}

EMF & FARADAY'S LAW

Transformer EMF *stationary loop,* $\mathbf{B}$ *changes*

$$V_{\text{emf}} = -\iint_S \frac{\partial\mathbf{B}}{\partial t}\cdot d\mathbf{s}$$

Motional EMF *loop moves,* $\mathbf{B}$ *static*

$$V_{\text{emf}} = \oint(\mathbf{v}\times\mathbf{B})\cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = -\frac{d\Phi}{dt}, \qquad \Phi = \iint_S \mathbf{B}\cdot d\mathbf{s}$$

N-turn coil

$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -\frac{d\lambda}{dt}, \qquad \lambda = N\Phi$$

Uniform B shortcut $\mathbf{B}$ *same everywhere on loop*

$$V_{\text{emf}} = -\frac{\partial B}{\partial t}\cdot A$$

Long wire + rect loop *wire near loop, r: $a\to b$, height c*

$$V_{\text{emf}} = -\int_0^c\int_a^b \frac{\partial}{\partial t}\left(\frac{\mu_0 I}{2\pi r}\right)dr\,dz$$

ROTATING LOOP — GENERATOR

$$\Phi = AB_0\cos(\omega t + \phi_0), \qquad V_{\text{emf}} = AB_0\omega\sin(\omega t + \phi_0)$$

$$V_0 = NA\omega B_0 \text{ (peak)}, \qquad \omega = 2\pi f = 2\pi\frac{\text{rpm}}{60}$$

2nd method: $V_{emf} = \oint(\mathbf{u}\times\mathbf{B})\cdot d\mathbf{l}$ on moving sides only.

IDEAL TRANSFORMER

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}, \qquad \frac{I_1}{I_2} = \frac{N_2}{N_1}, \qquad Z_{\text{in}} = \left(\frac{N_1}{N_2}\right)^2 Z_L$$

Step-up ($N_2 > N_1$): $V_2 > V_1$, $I_2 < I_1$. Power conserved: $V_1I_1 = V_2I_2$.

EQUIVALENT CIRCUIT (INDUCTOR IN CHANGING B)

Inductor (resistance R_i) in external changing $\mathbf{B}$*, connected to load R:*

$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i} \approx \frac{V_{\text{emf}}^{\text{tr}}}{R} \text{ when } R_i \ll R$$

V_{emf}^{tr} acts as a voltage source in series with R_i.

CROSS PROD CYCLE

$\hat{x}\times\hat{y} = \hat{z}$; $\hat{y}\times\hat{z} = \hat{x}$; $\hat{z}\times\hat{x} = \hat{y}$
$\hat{y}\times\hat{x} = -\hat{z}$; reverse: negate
$\mathbf{a}\times\mathbf{a} = 0$; $|\mathbf{a}\times\mathbf{b}| = ab\sin\theta$

B FIELD QUICK

Wire: $B = \mu_0 I/(2\pi r)$
Toroid: $B = \mu N I/(2\pi r)$
$\Phi = \iint \mathbf{B}\cdot d\mathbf{s}$ (Wb)
$V_{\text{emf}} = -N\,d\Phi/dt$

GENERATOR

$\Phi = AB_0\cos(\omega t + \phi_0)$
$V_{\text{emf}} = AB_0\omega\sin(\omega t + \phi_0)$
$V_0 = NA\omega B_0$ (peak)
$V^{\text{m}} = \oint(\mathbf{v}\times\mathbf{B})\cdot d\mathbf{l}$

j QUICK

$j^2 = -1$; $1/j = -j$
$e^{\pm j\pi/2} = \pm j$; $e^{j\pi} = -1$
$|a + jb| = \sqrt{a^2 + b^2}$
$\mathrm{Re}\{Ae^{j(\omega t+\theta)}\} = A\cos(\omega t + \theta)$

CONSTANTS

$\mu_0 = 4\pi\times 10^{-7}$ H/m
$\varepsilon_0 \approx \frac{1}{36\pi}\times 10^{-9}$ F/m
$c = 3\times 10^{8}$ m/s
$\eta_0 \approx 377 \approx 120\pi\ \Omega$

MAXWELL'S EQUATIONS

Law	Differential	Integral
Gauss $\mathbf{E}$	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint \mathbf{D} \cdot d\mathbf{s} = Q$
Gauss $\mathbf{B}$	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{s} = 0$
Faraday	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint \mathbf{H} \cdot d\mathbf{l} = I_c + I_d$

$\mathbf{D} = \varepsilon \mathbf{E}$; $\mathbf{B} = \mu \mathbf{H}$; $\mathbf{J} = \sigma \mathbf{E}$. *Phasor:* $\varepsilon_c = \varepsilon' - j\sigma/\omega$.

BOUNDARY CONDITIONS

Field	General	PEC ($\sigma \to \infty$)
Tang. $\mathbf{E}$	$E_{t1} = E_{t2}$	$E_t = 0$
Tang. $\mathbf{H}$	$H_{t1} - H_{t2} = J_s$	$\hat{n}_2 \times \mathbf{H} = \mathbf{J}_s$
Norm. $\mathbf{D}$	$D_{n1} - D_{n2} = \rho_s$	$D_n = \rho_s$
Norm. $\mathbf{B}$	$B_{n1} = B_{n2}$	$B_n = 0$

No J_s, no ρ_s: all four components continuous across boundary.

CONDUCTION vs DISPLACEMENT CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$E = \frac{V(t)}{d}, \quad J_c = \sigma E, \quad I_c = \frac{\sigma A}{d} V(t) = \frac{V}{R}, \quad R = \frac{d}{\sigma A}$$

Displacement *capacitive, changing E*

$$J_d = \varepsilon \frac{\partial E}{\partial t}, \quad I_d = \frac{\varepsilon_r \varepsilon_0 A}{d} \frac{dV}{dt} = C \frac{dV}{dt} \Rightarrow C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

Lossy capacitor ($R \parallel C$) *both σ and ε_r*

$$I = I_c + I_d = \frac{V}{R} + C\frac{dV}{dt}$$

Phasor: $\tilde{J}_d = j\omega\varepsilon \tilde{E}$. *In good conductor* $I_d \ll I_c$.

CHARGE RELAXATION

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \Rightarrow \oint \mathbf{J} \cdot d\mathbf{s} = 0 \text{ (KCL)}$$

$$\rho_v(t) = \rho_{v0} e^{-t/\tau_r}, \qquad \tau_r = \frac{\varepsilon}{\sigma} = \frac{\varepsilon_r \varepsilon_0}{\sigma} \text{ (s)}$$

Find σ from decay data

$$\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \varepsilon_r \varepsilon_0}{t_1}$$

Cu: $\tau_r \approx 10^{-19}$ *s. Mica:* $\tau_r \approx 10^4$ *s. Distilled water:* $\sim 10^{-6}$ *s.*

PHASOR METHOD ($\tilde{\mathbf{E}} \to \tilde{\mathbf{H}}$)

1. $\mathbf{E}(z,t) = \mathrm{Re}\{\tilde{\mathbf{E}}(z)e^{j\omega t}\}$ strip $e^{j\omega t}$, keep e^{-jkz}

2. Time → phasor:
$\cos(\omega t - kz) \to e^{-jkz}$; $\cos(\omega t - kz + \phi) \to e^{j\phi}e^{-jkz}$
$\sin(\omega t - kz) \to -je^{-jkz}$; $\partial/\partial t \to j\omega$

3. Faraday in phasor domain (plane wave: only ∂_z survives)

$$\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}$$

4. Phasor → time: $\mathbf{H}(z,t) = \mathrm{Re}\{\tilde{\mathbf{H}}e^{j\omega t}\}$
$\mathrm{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$
$\mathrm{Re}\{-je^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

DYNAMIC POTENTIALS

$$\mathbf{B} = \nabla \times \mathbf{A}, \qquad \mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t}$$

Statics: $\mathbf{E} = -\nabla V$. *Dynamic: must include* $-\partial \mathbf{A}/\partial t$ *term. Retarded: effects at* $\mathbf{R}$ *at time* t *depend on source at time* $t - R'/u_p$.

LOSSLESS PLANE WAVE ★ *(missing from CS1+CS2)*

Wavenumber & impedance

$$k = \omega\sqrt{\mu\varepsilon} \text{ (rad/m)}, \qquad \eta = \sqrt{\frac{\mu}{\varepsilon}} \; (\Omega)$$

Phase velocity & wavelength

$$u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} \text{ (m/s)}, \qquad \lambda = \frac{2\pi}{k} = \frac{u_p}{f} \text{ (m)}$$

Phasor fields (wave in $+\hat{z}$, E in $\hat{x}$)

$$\tilde{\mathbf{E}}(z) = E_{x0}^{+} e^{-jkz}\hat{x}, \qquad \tilde{\mathbf{H}}(z) = \frac{E_{x0}^{+}}{\eta} e^{-jkz}\hat{y}$$

Real fields

$$\mathbf{E}(z,t) = |E_{x0}^{+}| \cos(\omega t - kz + \phi^{+})\hat{x}$$

$$\mathbf{H}(z,t) = \frac{|E_{x0}^{+}|}{\eta} \cos(\omega t - kz + \phi^{+})\hat{y}$$

TEM: $\mathbf{E} \perp \mathbf{H} \perp \hat{k}$. *Lossless:* $\mathbf{E}$ *and* $\mathbf{H}$ *are* ***in phase***.

DIRECTION RELATION ★ *(missing from CS1+CS2)*

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{k} \times \tilde{\mathbf{E}}, \qquad \tilde{\mathbf{E}} = -\eta\,\hat{k} \times \tilde{\mathbf{H}}$$

$\hat{k}$	$\hat{E}$	$\hat{H} = \hat{k} \times \hat{E}$	Example $\tilde{\mathbf{E}}$
$+\hat{z}$	$+\hat{x}$	$+\hat{y}$	$E_0 e^{-jkz}\hat{x}$
$+\hat{z}$	$+\hat{y}$	$-\hat{x}$	$E_0 e^{-jkz}\hat{y}$
$-\hat{z}$	$+\hat{x}$	$-\hat{y}$	$E_0 e^{+jkz}\hat{x}$
$+\hat{x}$	$+\hat{y}$	$+\hat{z}$	$E_0 e^{-jkx}\hat{y}$

POLARIZATION ★ *(missing from CS1+CS2)*

General phasor *$a_x = |E_{x0}|$, $a_y = |E_{y0}|$, δ = phase E_y rel. E_x*

$$\tilde{\mathbf{E}} = (a_x\hat{x} + a_y e^{j\delta}\hat{y})e^{-jkz}$$

Type	Condition	Form
Linear	$\delta = 0, \pi$	$(a_x\hat{x} \pm a_y\hat{y})\cos(\omega t - kz)$
LHC	$a_x = a_y$, $\delta = +\pi/2$	$\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$
RHC	$a_x = a_y$, $\delta = -\pi/2$	$\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$
Elliptical	general a_x, a_y, δ	traces ellipse in x-y

LHC real field

$$\mathbf{E} = a\cos(\omega t - kz)\hat{x} - a\sin(\omega t - kz)\hat{y}$$

RHC real field

$$\mathbf{E} = a\cos(\omega t - kz)\hat{x} + a\sin(\omega t - kz)\hat{y}$$

Elliptical axial ratio *a_ξ=semi-major, a_η=semi-minor*

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta, \quad \sin 2\chi = (\sin 2\psi_0)\sin\delta$$

$$\psi_0 = \tan^{-1}(a_y/a_x), \quad R = a_\xi/a_\eta, \quad \tan\chi = \pm 1/R$$

$\tan\chi > 0$*: LH.* $\tan\chi < 0$*: RH.* $R = 1$*: circular.* $a_\eta = 0$*: linear.*

MEDIUM TYPES ★ *(missing from CS1+CS2)*

Type	$\varepsilon''/\varepsilon'$	Key approx.
Low-loss diel.	$\ll 1$	$\alpha \approx \frac{\sigma}{2}\sqrt{\mu/\varepsilon'}$, $\beta \approx \omega\sqrt{\mu\varepsilon'}$
Quasi-conductor	10^{-2}–10^2	use general α, β
Good conductor	$\gg 100$	$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$

$$\delta_s = \frac{1}{\alpha} \text{ (m)}, \quad \delta_s^{\text{cond}} = \frac{1}{\sqrt{\pi f \mu \sigma}} \text{ (m)}, \quad \varepsilon'' = \frac{\sigma}{\omega}$$

POYNTING & POWER

$$\mathbf{S}_{\text{av}} = \tfrac{1}{2}\,\mathrm{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\} \text{ (W/m}^2\text{)}$$

Lossless *$\alpha = 0$, η real*

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta}\hat{k}$$

Lossy *$\eta_c = |\eta_c|e^{j\theta_\eta}$, wave in $+\hat{z}$*

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z}\cos\theta_\eta\;\hat{z}$$

Power $\propto e^{-2\alpha z}$*, field* $\propto e^{-\alpha z}$*: power decays at* $2\times$ *rate.*

ATTENUATION & dB

$$\alpha \text{ (dB/m)} = 8.69\,\alpha \text{ (Np/m)}$$

$$A = 10\log\left(\frac{S(z)}{S(0)}\right) = -8.69\,\alpha z \text{ (dB)}$$

$$A = 20\log\left(\frac{|E(z)|}{|E(0)|}\right) \text{ (dB)}$$

$$G = 10\log\left(\frac{P_1}{P_2}\right) = 20\log\left(\frac{V_1}{V_2}\right) \text{ (dB)}$$

$P \times 2 \Rightarrow +3$ *dB;* $P \times 10 \Rightarrow +10$ *dB;* $|E| \times 2 \Rightarrow +6$ *dB.*

MAXWELL PHASORS

$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$
$\nabla \times \tilde{\mathbf{H}} = j\omega\varepsilon_c\tilde{\mathbf{E}}$
$\varepsilon_c = \varepsilon' - j\sigma/\omega$
$\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}$

WAVE PARAMS

$k = \omega\sqrt{\mu\varepsilon}$ (rad/m)
$\eta = \sqrt{\mu/\varepsilon}$ (Ω)
$u_p = 1/\sqrt{\mu\varepsilon}$ (m/s)
$\lambda = 2\pi/k = u_p/f$ (m)

POLARIZATION

Linear: $\delta = 0$ or π
LHC: $a_x = a_y$, $\delta = +\pi/2$
RHC: $a_x = a_y$, $\delta = -\pi/2$
Ellip.: general a_x, a_y, δ

MEDIUM + SKIN

$\varepsilon_c = \varepsilon' - j\sigma/\omega$
$\delta_s = 1/\alpha$ (general)
$\delta_s = 1/\sqrt{\pi f \mu \sigma}$ (cond.)
$R_s = \sqrt{\pi f \mu/\sigma}$ (Ω)

dB QUICK

$\alpha_{\text{dB}} = 8.69\,\alpha_{\text{Np}}$
$G = 10\log(P_1/P_2)$
$A = -8.69\alpha z$ (dB)
$A = 20\log(|E(z)|/|E(0)|)$